

The $3x + 1$ problem as interface physics: two substrates, quantized resonance, and the limits of proof

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Abstract

This is a discussion paper, but of an unusual kind: every structural claim in it is backed by a theorem, an exact identity, or an exhaustive computation from a companion corpus on the $3x+1$ map [1, 2, 3, 4, 5, 6]. We propose reading the map as the *minimal coupling of two substrates*: a 2-adic substrate in which halving is clean, and a 3-adic substrate in which the odd step is a pure digit append — the unique carry-free member of the $3n + d$ family. Each substrate alone is completely integrable; all difficulty is concentrated at the interface, whose coupling constant is the irrational relation $2^\alpha = 3$. From this reading four theses follow, each anchored in proof rather than analogy. (1) The map’s celebrated pseudo-randomness is a *theorem*, not an observation: refills are exactly fair, the macro-step law factorizes exactly, storage is exactly zero, and cross-base mutual information vanishes — randomness is what one substrate’s observer sees of clockwork coherent in the other, and noise is a relation between incommensurable receivers, not a property of the signal. (2) Cross-substrate coherence is not impossible but *quantized*: cycles are standing waves living only on the ladder of continued-fraction convergents of $\log_2 3$ (Arnold tongues of the 2:3 coupling), a correspondence made flesh by the verified cycle zoo of the $3n + d$ family ($3n + 13$, sitting on the Pythagorean limma $256/243$, has exactly seven free 5-step cycles). (3) Wolfram’s irreducibility thesis must be refined in both directions: the ensemble level is *more* reducible than his picture suggests (exactly solvable), while the pointwise level resists all reduction *provably*; moreover the zero-storage results weigh against the standard (Conway-style) mechanism for undecidability of this instance. (4) The two failure modes of the conjecture obey opposite ledgers — cycles are localized and spectral, divergence is delocalized and diffuse — and this asymmetry is mirrored in logic: membership in the tree of 1 has short certificates of length $O(\log n)$, while non-membership is not even semi-decidable by computation. The hypothetical “infinite winner” demanded by compactness intuition does exist — but in $\mathbb{Z}_2$, not in $\mathbb{Z}$: the conjecture is exactly the statement that the limit always escapes the integers. We close with an honest ledger of what is proved, what is structural template, and the three walls that remain.

1 The two-substrate reading

1.1 A map with two homes

The Syracuse form of the Collatz map sends an odd integer n to $T(n) = (3n + 1)/2^{v_2(3n+1)}$. Sixty years of work have treated this as one dynamical system on one space. The reading we develop here — and which the companion corpus turns from metaphor into mathematics — is that it is better seen as *two* exactly solvable systems and one hard *interface*.

Write an integer simultaneously in base 2 and in base 3. The two halves of the map then live in different worlds:

- **The 2-adic substrate.** Halving is a pure shift in binary: the binary digit sum s_2 is invariant, and the branching decision (how many halvings w follow an odd step) is read off the low binary bits. The map’s entire control flow — the parity word — is 2-adic. By the Terras correspondence, depth- r index patterns occupy exactly one residue class modulo $2^{\text{bits used}}$, so pattern densities are *exact*, not asymptotic [Note, Thm. 10], [21].
- **The 3-adic substrate.** The odd step is invisible complexity in base 2 (a multiplication with carries) but trivial in base 3: $3r + 1$ *appends the trit* 1, with zero propagation into the digits of r , so the ternary digit sum obeys $s_3 \mapsto s_3 + 1$ exactly [Note, Thm. 25]. Halving in base 3 is likewise an exact automaton: a top-down parity sweep with a six-entry shed table [Note, Thm. 36], and even the branch condition closes inside base 3, since $x \bmod 2^j$ is an explicit periodic-weight trit sum [Note, Lem. 37].

So each substrate, taken alone, is clean: the map is an odometer in one world and a clockwork in the other. Where, then, does sixty years of difficulty live?

1.2 Each substrate alone is exactly solvable — and therefore empty

The corpus contains two no-go results, one per substrate, and together they answer the question with unusual precision.

On the 2-adic side: *no function of the residue $n \bmod 2^j$ can be a strict Lyapunov certificate at any depth j* [Note, Thm. 11]. The proof is the sharpness of the shadowing bound: the residue class of $-5 \bmod 2^j$ admits, at every depth, a consistent macro-step continuation returning to the class of -5 with value multiplied by $9/8 > 1$. The negative cycles of the map are proof obstructions for the whole class of congruence-state methods; any certificate must use archimedean information.

On the 3-adic side: after an entire campaign that produced the exact division automaton and a family of exact conservation laws — the top tax [Note, Thm. 40], the zeros ledger [Note, Thm. 41], run pairing [Note, Thm. 44], the slot lifecycle [Note, Thm. 43] — the sector was proved *completely integrable*: every static ledger and every digit-sum currency in the append/sweep automaton closes tautologically by conservation, and no Lyapunov function exists in this sector either [Note, Prop. 45].

This pair of results deserves to be stated as the structural heart of the two-substrate reading:

Principle 1.1 (Integrable substrates, hard interface). Each substrate of the $3x + 1$ map is completely integrable, and precisely for that reason carries no proof: its conservation laws close on themselves. All content of the conjecture lives at the interface between the substrates, whose coupling constant is the irrational relation $2^\alpha = 3$ (i.e. $\alpha = \log_2 3 \notin \mathbb{Q}$).

The phrase “coupling constant” is not decorative. The identity $2^\alpha = 3$ is literally what pins the interface to criticality: at the edge of the Krasikov–Lagarias system the average row mass is *exactly* 1 because $3/4 + 3 \cdot 2^{\alpha-2} = 3$ [Note, Lem. 24], and the offset-mass flows up and down the 3-adic scale ladder balance exactly, $3/16 = 3/16$ [Note, Prop. 32(i)]. The defining equation of the problem is precisely the statement that the system sits on the conservative line: neither substrate wins for free.

1.3 Minimality: $3n + 1$ is the weakest coupling that still couples

Within the family $3n + d$ (d odd), the choice $d = 1$ is distinguished twice over, and both distinctions are theorems.

First, the carry characterization: $3n + 1$ is the *unique* base-3-local member of the family. For every other offset, $3r + d$ carries or borrows into the digits of r with geometrically distributed depth (for $3r - 1$, the borrow depth law is exactly $P(\text{depth} = j) = 2/3^j$) [Note, Thm. 25]. Only $d = 1$ touches the ternary clockwork with a bare append — the minimal possible contact.

Second, the pair law: consecutive segment endpoints within a family satisfy $x' = 3x + 2$, hence $x'/2 = 3(x/2) + 1$; the guaranteed merge structure of the Collatz graph exists for the multiplier 3 and for *no other* map $cn + 1$ [Note, Thm. 2].

The convergent map is thus the minimally coupled and maximally local member of its family. This inverts the usual intuition that $3n + 1$ is a generic representative made hard by accident: it is the *extremal* member, the weakest coupling that still connects the two substrates — and, empirically and conjecturally, the one that converges. The reading suggested by the whole corpus is that these two facts are the same fact.

2 Randomness from determinism as a theorem

2.1 Three grades of deterministic pseudo-randomness

Wolfram’s *A New Kind of Science* [22] made Rule 30 the emblem of randomness generated by simple deterministic rules. But the epistemic status of that randomness is empirical: the center column of Rule 30 passes statistical tests, and no proof of any strong randomness property is known. Rule 110 sits differently: its interesting property — universality — is proved [11], and its randomness is incidental. The $3x + 1$ map, we claim, occupies a third grade, strictly above both for this purpose: *its randomness is a theorem*. Specifically:

- **Exact fairness.** The set of odd n whose first r successive ladder refills (trailing-ones depths after each full burn) are all $\geq j$ has density *exactly* $(2^{-(j-1)})^r$: successive refills are exactly independent fair geometric draws at density level [Note, Thm. 20].
- **Exact factorization.** Exhaustively over odd $n < 2^{22}$, the within-step joint law of (trailing ones k , post-burn halvings w) factorizes exactly: $I(k; w) = 0.000000$ bits and $P(w \mid k) = 2^{-w}$ for every k , so $P(k, w) = 2^{-k}2^{-w}$ [Note, Thm. 27].
- **Zero storage.** The mutual information between refill depths along macro-orbits, $I(k_0; k_d)$ for $d = 1, 2, 3$, is 0 to within 3×10^{-6} bits against an entropy of 1.9375 bits: the macro-automaton has zero channel capacity at density level [Note, Thm. 22].
- **Cross-base blindness.** The apparent coupling $I(n \bmod 9; w_{\text{next}})$ vanishes as the 2-adic modulus grows: it was finite-modulus leakage, and the true cross-base mutual information is zero [Note, Thm. 27].

These are counted quantities, not estimated ones. The density-level stochastic model of the map is *exactly* independent fair geometric dice with no measurable correction at any level; every hiding place for density-level structure is closed [Note, Thm. 27; Forgetful].

2.2 The mechanism: noise as inter-receiver incommensurability

What makes this more than a curiosity is that the corpus also identifies the *mechanism*, and the mechanism is the interface of Section 1. The “randomness” seen in the 2-adic frame is the 2-adic reading of structure that is perfectly coherent in the 3-adic frame: an append-only clockwork [Note, Thm. 25], driven at the top by a Sturmian rotation governed by the irrationality of $\log_2 3$

[Note, Prop. 38]. Below the interface sits a rational odometer (perfectly periodic bit-clocks); above it, an irrational rotation; their product, read in either single-base frame, is exact fair dice. The incommensurability of the coupling constant $2^\alpha = 3$ is the noise source.

This suggests a definition that we state as a proposal, because its mathematical content here is proved while its generality is not:

Principle 2.1 (Noise as a relation). Noise is not a property of a signal; it is a relation between receivers. Determinism in substrate A appears as randomness in receiver B exactly when their bases are incommensurable. In the $3x + 1$ instance this is quantitative: the fairness is exact at ensemble level [Note, Thms. 20, 27], and the anti-storage theorem [Note, Thm. 22] shows no bounded single-base receiver can exploit the determinism underneath — there is no channel — while the full two-base observer sees pure clockwork.

Under this reading the conjecture itself acquires a striking form: even the orbit’s *own thread* cannot exploit its determinism against the interface. Convergence is self-decoherence.

3 Quantized resonance: cycles on the comma ladder

3.1 Not “no coherence” but “spectral coherence”

Section 2 might suggest that cross-substrate coherence is simply impossible: exact fair dice, zero storage, zero mutual information. But that would overstate what is proved, and the overstatement would miss the most beautiful structure in the problem. What the theorems exclude is *generic*, broadband coherence: a typical signal cannot hold both receivers. Coherent objects nonetheless exist — cycles. A cycle is a standing wave at the interface: periodic, hence coherent in the 2-adic frame (its parity word repeats) and in the 3-adic frame (its trit-stream statistics repeat), forever.

Where can such standing waves live? The cycle equation for a shape with k odd steps and s total halvings reads

$$n(2^s - 3^k) = B(\text{shape}), \quad B > 0 \text{ a sum of positive terms,}$$

so an integer cycle forces $2^s - 3^k$ to be small and positive relative to B : the ratio s/k must be a near-best rational approximation of $\alpha = \log_2 3$ [Note, Thm. 5]. The permitted sites are the continued-fraction convergents of $\log_2 3$ — in musical language, the ladder of *commas*: $2^2/3^1$ (the touch $4 - 3 = 1$), $2^5/3^3 = 32/27$, $2^8/3^5 = 256/243$ (the Pythagorean limma), $2^{19}/3^{12}$ (the comma proper), and so on. Cycles are not forbidden; they are *quantized*. The dynamical-systems name for this phenomenon is mode-locking: the comma ladder is the Arnold-tongue spectrum [7] of the 2:3 coupling.

3.2 The comma-cycle correspondence, made flesh

If this spectral picture is right, it should be *constructive*: sitting a map exactly on a rung of the ladder should produce cycles in abundance. It does, and this is a theorem with a verified zoo attached. For each near-touch pair $(3^k, 2^s)$ with gap $d = 2^s - 3^k > 0$, the map $n \mapsto (3n + d)/2^w$ has *free* cycles with exactly k odd steps and s halvings: the cycle equation $n(2^s - 3^k) = dB$ cancels to $n = B$, so every composition shape yields a cycle, subject only to parity — $\binom{s-1}{k-1}$ shapes per rung [Note, Prop. 48]. Verified instances:

map	rung of the ladder	gap d	free cycles (odd starters)
$3n + 7$	$2^4/3^2 = 16/9$	7	5, 7, 11
$3n + 5$	$2^5/3^3 = 32/27$	5	19, 23, 29, 31
$3n + 13$	$2^8/3^5 = 256/243$ (limma)	13	211, 227, 251, 259, 283, 287, 319

The map $3n + 13$, sitting on the limma, has exactly *seven* distinct free 5-step cycles; explicitly, $211 \rightarrow 323 \rightarrow 491 \rightarrow 743 \rightarrow 1121 \rightarrow 211$. The notorious cycle zoo of the $3n + d$ family is not a zoo at all: it is the comma ladder made flesh, each convergent of $\log_2 3$ endowing its gap value d with a full resonance family.

3.3 Why $3n + 1$ is poor: the free resonances are exhausted

Conversely, the cycle *scarcity* of $3n + 1$ is the statement that $d = 1$ sits only on the exact touches: integrality in the cycle equation is automatic iff $|2^s - 3^k| = 1$. By Mihăilescu’s theorem (Catalan’s conjecture) [18] the difference-1 pairs are exhausted by $(k, s) = (1, 2)$ and $(2, 3)$: exactly the trivial cycle $\{1, 4, 2\}$ and the -5 cycle. The two free cycles arithmetic can give, it has given [Note, Thm. 5, Cor.].

For the cases actually needed, full Catalan is a luxury: elementary congruences suffice, and the observation is old. Modulo 8, $3^m + 1$ is 2 or 4, never 0, so $2^n = 3^m + 1$ has no solution with $n \geq 3$ — one line kills all infinitely many candidates at once [Note, §12]. And $3^m = 2^n + 1$ forces m even, whence $(3^{m/2} - 1)(3^{m/2} + 1) = 2^n$ gives $9 = 8 + 1$ only. This last case — that $(8, 9)$ is the final pair of consecutive 3-smooth numbers — was proved by Levi ben Gershon (Gersonides) in 1343, in *De numeris harmonicis*, answering a question of the composer Philippe de Vitry about which pairs of harmonic numbers can differ by one [12]. The exhaustion of the free resonances of the 2:3 interface is, fittingly, a result born in music theory — the same ladder that governs why a circle of twelve fifths misses seven octaves by a comma.

The ground state $\{1, 4, 2\}$ is itself a resonance, living on the touch $4 - 3 = 1$. The conjecture is therefore *not* “there is no resonance” — that is false, the trivial cycle is one — but:

Principle 3.1 (The conjecture as a ground-state statement). The ground resonance is globally attracting: every orbit decoheres into the one permitted mode. Between incommensurable substrates, coherence exists only on the discrete comma spectrum of the coupling constant; broadband coherence is impossible; and released systems drain into the lowest available resonance.

3.4 The same ladder governs the stopping-time structure

That the comma ladder is the problem’s true skeleton — not merely its cycle bookkeeping — is confirmed from an independent direction. Terras’ coefficient-stopping-time conjecture (that $\tau(n) = \sigma(n)$ for $n > 1$) was verified exhaustively for all stopping classes with $\tau \leq 24$ (81,119 classes) and all odd $1 < n \leq 10^7$; the smallest safety margin observed is 2.02, and the extremal classes concentrate at $(u, j) = (5, 8)$ — the limma again [Note, Props. 28, 29]. A CST violation in class (u, j) requires $n \leq b/(2^j - 3^u)$, which explodes exactly at the convergents of $\log_2 3$; with the five-line bound $b/2^j < u/3$ the reduction becomes unconditional and extends CST to $\tau \leq 4700$ [CstComma, Cor. 6.2]. Even the growth of the extremal numerators is comma-governed: $B(t) \sim 1 + t/8$, one unit per repetition of the limma word $(5, 8)$ [Note, Lem. 30]. Cycle exclusion and CST are one wall seen from two sides: effective rational approximation of $\log_2 3$ — Baker territory [8, 19].

4 Wolfram’s irreducibility thesis, refined

Wolfram presents the Collatz problem as a poster child of computational irreducibility, with the suggestion — backed by Conway’s universality theorem for the generalized class [10] — that the instance may be undecidable [22]. The corpus forces a refinement of this picture in *both* directions at once, and the refinement is more interesting than either verdict alone.

4.1 The ensemble level is more reducible than Wolfram’s picture

Irreducibility, in Wolfram’s usage, admits “pockets of reducibility.” What the corpus shows is stronger: at ensemble (density) level the $3x + 1$ map is not a pocket but *exactly solvable*. The stationary roulette measure has closed form (mod 9: $\pi(1, 2, 4, 5, 7, 8) = (8, 16, 11, 4, 2, 22)/63$); refills are exactly fair [Note, Thm. 20]; the macro-step law factorizes exactly [Note, Thm. 27]; storage is exactly zero [Note, Thm. 22]; the linearized Krasikov–Lagarias edge operator has spectrum $\{1\} \cup \{|\lambda| = 1/4\}$ exactly [Note, Thm. 21]; and the Perron eigenvectors of the certificate system are, to $R^2 > 0.99$, pure powers of the roulette measure [Tempering]. In the language of coarse-graining: lumpability of the macro-chain is exact — a theorem here, not an approximation. Tao’s celebrated “almost all orbits attain almost bounded values” [20] is, in this reading, exactly what coarse-graining success at ensemble level looks like from the analytic side.

4.2 The pointwise level resists all reduction, provably

At the level of the individual orbit the resistance is not an impression but a set of impossibility results: no residue-based Lyapunov certificate at any 2-adic depth [Note, Thm. 11]; complete integrability (hence certificate-emptiness) of the entire digit-sum sector [Note, Prop. 45]; mirror-blindness — the Krasikov–Lagarias growth exponent is identical for $3n - 1$ and $3n + 1$, so density methods cannot even distinguish the convergent map from a map with three known nontrivial cycles [Forgetful]; and the no-go on coarse resampling — the cascade attenuation κ is not a functional of any coarse histogram; only exact structure reproduces it [Note, Campaign XVIII]. The reducible/irreducible boundary is thus not fuzzy but *located*: it is the ensemble/pointwise split, and it is proved from both sides.

4.3 Against the hidden computer

Wolfram’s suspicion of undecidability inherits its plausibility from Conway: generalized Collatz maps can simulate arbitrary register machines, encoding registers in the exponents of primes, and the generalized problem is Π_2^0 -complete [10, 15]. But every known construction of this type works by *storing information*. The specific $3n + 1$ instance exhibits the opposite signature on every test: zero channel capacity at density level [Note, Thm. 22], exact memorylessness [Note, Thms. 20, 27], and the carry characterization — the unique append-only, maximally forgetful member of its family [Note, Thm. 25]. This is the signature of an odometer, not of a Turing machine [Forgetful]. Undecidability of the instance remains possible; what the evidence weighs against is the *standard mechanism* for it. If the $3x + 1$ problem is independent of strong theories, it will not be because a computer hides in it, but for a reason not yet named — which would itself be a discovery about the reach of the Conway paradigm.

5 The two requirement ledgers and the semi-decidability asymmetry

The conjecture has exactly two failure modes: a nontrivial cycle, or a divergent orbit. The corpus compiles, from proved results only, what each would have to satisfy — and the two ledgers turn out to have *opposite shapes*.

5.1 The cycle ledger: localized and spectral

A hypothetical cycle with k odd steps and s halvings must, simultaneously [Note, §11 of the synthesis; Thms. 5, 9, 17]:

1. live on the comma ladder: $n = B/(2^s - 3^k) \in \mathbb{Z}$ forces $s/k \approx \log_2 3$ (near-touch), so its possible homes are spectrally enumerated;
2. survive the exhaustion of the free homes: gap-1 sites carry only the trivial cycle (elementary today; Gersonides 1343 for the 9/8 case; Mihăilescu in general), so it needs the divisibility miracle $(2^s - 3^k) \mid B(\text{shape})$ against superpolynomially growing gaps;
3. have a net-falling shape, by the Sign Theorem: every net-climbing shape realizes at a *negative* 2-adic point [Note, Thm. 5];
4. balance exact periodic ledgers: per period the digit-sum shed equals u exactly and the top tax equals j , so its lower sweep must gain exactly $j - u$ per period [Note, Thm. 40];
5. be large: no integer cycles with ≤ 14 odd steps beyond the known four $\{1, -1, -5, -17\}$ [Note, Thm. 17, extended]; at least 1.375×10^{11} odd terms and ≥ 92 circuits by Hercher's bound [13]; all elements above 2^{71} by exhaustive verification [9].

This ledger is *localized*: the candidate sites are known, discrete, and individually checkable; finitely many checks per site plus one effective gap bound (Baker-type) would finish cycles off entirely. A cycle, if it exists, is a standing wave on a named Arnold tongue — we know exactly where it could live.

5.2 The divergence ledger: delocalized and diffuse

A hypothetical divergent orbit must [Note, §11; Thms. 9, 9', 10, 20, 40; Prop. 13]:

1. hold $\bar{w} < \log_2 3$ forever — a sustained excess of $w = 1$ steps against dice proved exactly fair, an event of measure zero in every window, required pointwise [Note, Thm. 20];
2. have an aperiodic parity word (else it is a cycle), and moreover escape *every* periodic and eventually-periodic index pattern within logarithmically many digits: shadowing any non-realized rational phase is throttled at $r \leq \log_2(n + |\rho|)/d$ periods, unconditionally [Note, Thms. 9, 9'] — divergence requires unbounded aperiodic complexity, quantitatively;
3. pay for every monotone climb at the exact exponential price μ^r with $\mu = 0.28627450\dots$ [Note, Thm. 10];
4. pay the top tax — an exact +1 shed at the top of every halving, pointwise, every orbit [Note, Thm. 40] — while its length grows;
5. and yet need *no arithmetic coincidence at all*: no divisibility miracle, no special site. Its requirement is diffuse — it must win everywhere, forever.

Divergent orbits are excluded to density $x^{0.9146}$ (certified at 3-adic depth $k = 20$ [Tempering]) and below 2^{71} [9]; what survives is a measure-zero possibility with no location. The asymmetry is stark: *cycles are localized and spectral; divergence is delocalized and diffuse*. The wall protecting cycles is transcendence theory; the wall protecting divergence is the pointwise/measure-zero gap itself.

5.3 The logical mirror: membership confesses, non-membership never

This dynamical asymmetry has an exact logical shadow. Membership in the tree of 1 is semi-decidable with *short* certificates: if n reaches 1, its own orbit is the proof, and its length is at most $c^*(n) \log_2 n$ with the laggard-edge constant $c^* \approx 18$ at 2^{24} , growing slowly toward ≈ 29 — a polynomial-size certificate [Note, §13]. Every number ever named has been proven in; there are no known hard numbers, only numbers nobody has run. If the conjecture is true, *every* integer has a short membership proof.

Non-membership is worse than hard: it is not even semi-decidable by running, because a divergent orbit never confesses. No individual integer can ever be proven *out* by computation alone. Uniformly, the residue classes lacking a depth- k proof are exactly the survivor classes mod 2^k — counting $\sim 1.834^k$, density $0.917^k \rightarrow 0$, ones-rich (the kin of 63 mod 64, the shadow set of the rational phases $-1, -5, -17$, mean ones-fraction 0.653) [Note, §13]. At proof budget t the unproven region is exactly the wavefront $\log_2 n > t/c^*$: the tree is provable number by number at logarithmic cost each; what cannot be bought at any finite budget is the single sentence *and this works for all of them*. This is the Π_2^0 shape of the problem [15] made concrete: a Σ_1 membership predicate with short witnesses, universally quantified.

6 Where the infinite winner lives

There is a persistent intuition — every student of the problem has felt it — that runs: *there are infinitely many integers; the conditions for escape are just statistics; surely some number satisfies all of them*. The corpus lets us answer this intuition precisely, and the answer has three layers, of which the third is the deepest fact in this essay.

(a) For cycles, the conditions are equations, not lottery tickets. Infinitely many candidates are powerless against a modular obstruction: there are infinitely many integers, and none of them satisfies $2^n = 3^m + 1$ beyond $4 = 3 + 1$ — one congruence mod 8 kills them all at once [Note, §12]. For the remaining candidate sites, the expected number of divisibility miracles is $\sum_{\text{sites}} O(1/\text{gap})$, which *converges*: infinity supplies candidates logarithmically, the conditions cut exponentially.

(b) For divergence, the heuristic argues for the conjecture. Per candidate n , the model probability of beating the exactly fair coin forever is the limit of 2^{-cT} , which is 0; countable additivity over infinitely many candidates gives expected winners = 0. Infinity times zero is zero here, not a loophole.

(c) And yet the winner exists — one substrate up. At every finite depth T , integers satisfying the first T escape conditions *do* exist: the survivor classes are never empty, numbering $S(T) \sim 1.834^T$ [Note, §13]. But they are *different* integers at each depth, running off to infinity, and the sequence of satisfiers converges — in $\mathbb{Z}_2$, not in $\mathbb{Z}$. The pure-climb example is exact: the satisfier of the k -fold climb condition is $2^k - 1$, and as $k \rightarrow \infty$,

$$2^k - 1 \longrightarrow -1 = \dots 111_2 \in \mathbb{Z}_2,$$

a genuine fixed point of pure climbing — and not a positive integer. In the 2-adic completion divergent orbits are not rare but *abundant*: almost every 2-adic point diverges. The compactness

winner your intuition demands exists as a pseudo-orbit: real in one receiver, noise in the other, finite in one substrate only.

This is where the two-substrate reading closes its loop. The integers are precisely the objects that are *finite in both substrates simultaneously* — finite binary top and finite ternary top; 2-adic pseudo-orbits are finite in one receiver only, and the archimedean property is exactly dual finiteness [Note, §3 of the synthesis]. Descent is *powered* by the finite top: the digit-sum drainage that pays the conservation ledger is produced almost entirely at the top boundary (+0.485 per trit in the top decile, against ≈ 0 in the bulk), the one feature integers have and pseudo-orbits lack [Note, Props. 38, 39, Thm. 40]. Hence:

Principle 6.1 (The conjecture as an escape statement). The survivor construction always produces its winner — in $\mathbb{Z}_2$. The Collatz conjecture is exactly the statement that the limit always escapes the integers: no dually-finite point realizes the infinite climbing stream. The infinite winner demanded by compactness intuition lives one substrate up, outside the countable thread.

7 Honest closing: proved, template, and the three walls

A synthesis essay owes its reader an exact accounting of epistemic status. We close with three lists.

7.1 What is proved or exhaustively computed

Every load-bearing claim above is anchored: the carry characterization and uniqueness of $3n + 1$ [Note, Thm. 25]; the pair-law uniqueness of the multiplier 3 [Note, Thm. 2]; the two substrate no-gos [Note, Thm. 11, Prop. 45]; exact fairness, factorization, and zero storage [Note, Thms. 20, 22, 27]; the conservative-line identities pinned by $2^\alpha = 3$ [Note, Lem. 24, Prop. 32]; the Sign Theorem and the Catalan corollary [Note, Thm. 5]; the cycle census through period 14 [Note, Thm. 17]; the shadowing bounds [Note, Thms. 9, 9']; the comma-cycle correspondence with its verified zoo [Note, Prop. 48]; the CST reduction to the convergents of $\log_2 3$, unconditional to $\tau \leq 4700$ [CstComma]; the certified density exponent $\gamma = 0.9146$ [Tempering]; and the exact edge rate $d\gamma/dq = 1/\ln(4/3)$ [Note, Thm. 19]. Where the corpus retracted — the per-position frontier ledger, the flagged $\bar{w} \leq 1.226$ bound — the retractions are recorded in the same documents [Note, corr. to Thm. 42, Prop. 45], and nothing in this essay leans on them.

7.2 What is structural template, not theorem

The physics vocabulary — receivers, coherence, decoherence, resonance, mode-locking — is proved *about the 2:3 arithmetic interface* and about nothing else. The proposal that noise in general is inter-receiver incommensurability (Principle 2.1), and any transfer of the mode-locking template to interfaces outside arithmetic, is a structural analogy whose merit is that it is falsifiable in form: a claimed instance must name its coupling constant, its convergent, and its tongue width. The mathematics here cannot say whether such transfers are deep or merely well-matched; we flag the boundary and stay on the proved side of it.

7.3 The three walls

What remains of the $3x + 1$ problem, after the corpus, is exactly three walls — and naming them precisely is the essay's final claim.

1. **The transcendence wall** (cycles, and CST). Both cycle exclusion and the coefficient-stopping-time conjecture reduce to effective lower bounds on $|2^j - 3^u|$ at the convergents of $\log_2 3$ [Note, Prop. 28, Thm. 29]: one wall, two classical sub-problems leaning on it. Current effective irrationality measures [8, 19] push the danger zone out (first convergent escape at $(u, j) = (15601, 24727)$ [CstComma]) but do not close it.
2. **The pointwise wall** (divergence). Every density-level statement is now exact, and exactly fair — and therefore exactly powerless per orbit: the pointwise version of Theorem 20’s fairness *is* the remaining content of the divergence half [Note, Thm. 20]. The sharpest exact formulation is the automaton form: no finite-top trit stream can sustain the self-referential (c, d) -statistics that perpetual climb requires [Note, Thms. 36, 40, Lem. 37] — the archimedean bridge, still unbuilt.
3. **The contraction wall** (density $\gamma \rightarrow 1$). The Krasikov–Lagarias program has been reduced to the stability of a one-dimensional fixed point: prove the cascade attenuation κ stays uniformly below 1 (measured 0.839 ± 0.002 , flat in depth) and $\gamma \rightarrow 1$ follows through proved identities [Note, Thm. 19, Props. 35, 47; Drift; GammaOne]. This is the smallest formulation the density problem has ever had — and even fully won, it buys $x^{1-\epsilon}$, not the pointwise sentence.

The three walls are not equally shaped: the first is localized (finitely many sites plus one effective bound), the third is a fixed-point stability statement with every coefficient measured, and the second is the conjecture’s irreducible core — the place where, on the evidence assembled here, the ensemble world ends and the individual thread begins. The two-substrate reading does not breach that wall. Its contribution is to show exactly where the wall stands, why everything on one side of it is clockwork, and why the one sentence on the other side — *every dually-finite signal decoheres to the ground state* — is the whole of the $3x + 1$ problem.

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This essay synthesizes results obtained over roughly 1200 rounds of human–AI collaborative research; the division of labour is recorded in the companion papers. The framing questions of Sections 2, 3, and 6 — whether the randomness has a mechanism, whether multi-layer resonance is impossible or quantized, and where the infinite winner lives — are due to the first author.

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